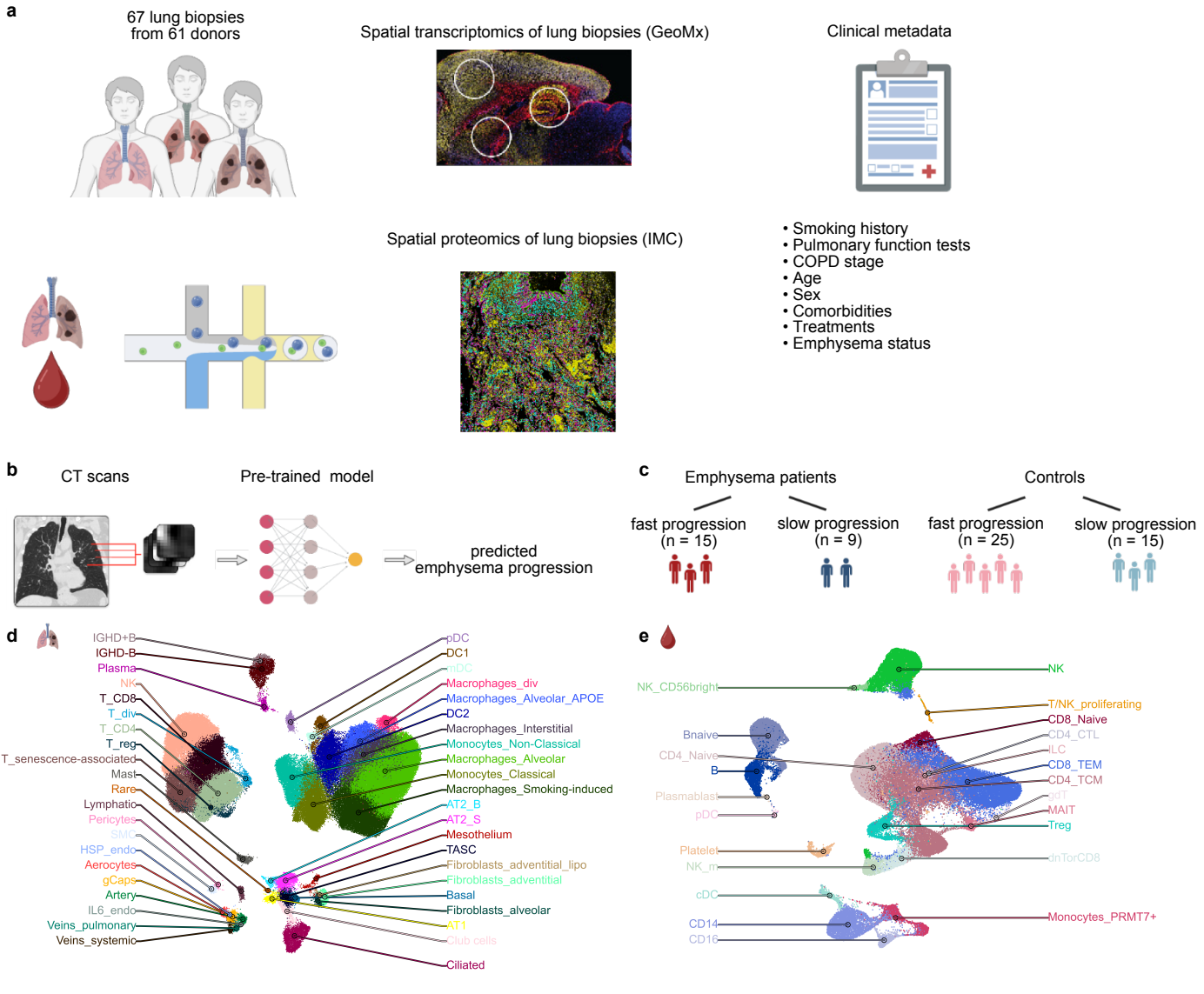

a
67 lung biopsies
from 61 donors
Spatial transcriptomics of lung biopsies (GeoMx)
Clinical metadata
Spatial proteomics of lung biopsies (IMC)
• Smoking history
• Pulmonary function tests
• COPD stage
• Age
• Sex
• Comorbidities
• Treatments
• Emphysema status
b
CT scans
Pre-trained model
predicted
emphysema progression
c
Emphysema patients
Controls
fast progression
(n = 15)
slow progression
(n = 9)
fast progression
(n = 25)
slow progression
(n = 15)
d
IGHD+B
IGHD-B
Plasma
NK
T_CD8
T_div
T_CD4
T_reg
T_senescence-associated
Mast
Rare
Lymphatic
Pericytes
SMC
HSP_endo
Aerocytes
gCaps
Artery
IL6_endo
Veins_pulmonary
Veins_systemic
pDC
DC1
nDC
Macrophages_div
Macrophages_Alveolar_APOE
DC2
Macrophages_Interstitial
Monocytes_Non-Classical
Macrophages_Alveolar
Monocytes_Classical
Macrophages_Smoking-induced
AT2_B
AT2_S
Mesothelium
TASC
Fibroblasts_adventitial_lipo
Fibroblasts_adventitial
Basal
Fibroblasts_alveolar
AT1
Club cells
Ciliated
e
NK_CD56bright
Bnaive
CD4_Naive
B
Plasmablast
pDC
Platelet
NK_m
cDC
CD14
CD16
NK
T/NK_proliferating
CD8_Naive
CD4_CTL
ILC
CD8_TEM
CD4_TCM
gdT
MAIT
Treg
nTorCD8
Monocytes_PRMT7+